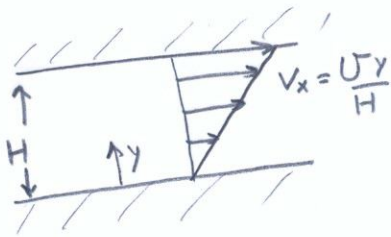


Consider the steady flow of a liquid confined between two parallel planar walls separated a distance H . If the top wall moves with constant velocity U , the resulting fluid motion exhibits a single velocity component $v_x = Uy/H$, where y denotes the distance to the bottom wall, assumed to be at rest.

1. Determine the streamlines and the trajectories
2. Obtain the vorticity field
3. Compute the circulation Γ around a rectangular line of side L oriented parallel to the walls and height h .
4. Calculate the gradient of velocity, the rate-of-strain tensor and the expansion rate.
5. Obtain the principal directions of strain as well as the associated principal strain rates.
6. Investigate the evolution of a square fluid element of side dl oriented parallel to the walls.



$$1) \frac{dx}{v_x} = \frac{dy}{v_y} \rightarrow \boxed{y = y_0} \rightarrow \text{STREAMLINES}$$

$$\left. \begin{aligned} \frac{dx}{dt} &= \frac{Uy}{H} \quad \boxed{x - x_0 = \frac{Uy_0}{H}t} \\ \frac{dy}{dt} &= 0 \rightarrow \boxed{y = y_0} \end{aligned} \right\} \text{TRAJECTORIES (PATHLINES } y = y_0)$$

$$2) \bar{\omega} = \nabla \wedge \bar{v} = \omega_z \bar{e}_z = \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} \right) \bar{e}_z = -\frac{U}{H} \bar{e}_z$$

3)

$$\Gamma = \oint \bar{v} \cdot d\bar{r} = \int_x^{x+L} v_x(y) dx + \int_y^{y+h} v_y(x) dy - \int_x^{x+L} v_x(y+h) dx - \int_y^{y+h} v_y(x) dy = \frac{UyL}{H} - \frac{U(y+h)L}{H} = -\frac{UhL}{H}$$

ALTERNATIVELY: $\Gamma = \int \bar{\omega} \cdot \bar{e}_z d\sigma = -Lh \frac{U}{H}$

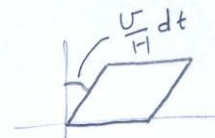
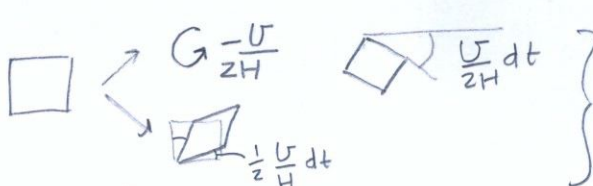
$$4) \nabla \bar{v} = \begin{bmatrix} 0 & 0 \\ \frac{U}{H} & 0 \end{bmatrix}, \bar{T}_d = \begin{bmatrix} 0 & \frac{1}{2} \frac{U}{H} \\ \frac{1}{2} \frac{U}{H} & 0 \end{bmatrix}, \bar{T}_r = \begin{bmatrix} 0 & -\frac{1}{2} \frac{U}{H} \\ \frac{1}{2} \frac{U}{H} & 0 \end{bmatrix}, \nabla \cdot \bar{v} = 0$$

$$5) |\bar{T}_d - \lambda \bar{I}| = 0 \quad \lambda^2 = \left(\frac{1}{2} \frac{U}{H} \right)^2 \rightarrow \lambda = \pm \frac{1}{2} \frac{U}{H} \quad \lambda_1 = +\frac{1}{2} \frac{U}{H}, \bar{n}_1 = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right)$$

$$\lambda_2 = -\frac{1}{2} \frac{U}{H}, \bar{n}_2 = \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \right)$$

$$6) \underline{dl(1,0)} \rightarrow d\ell(1,0) \quad \underline{dl(0,1)} \rightarrow d\ell(0,1) + d\ell(0,1) \begin{bmatrix} 0 & 0 \\ \frac{U}{H} & 0 \end{bmatrix} dt = d\ell\left(\frac{U}{H} dt, 1\right)$$

$$+ d\ell(1,0) \begin{bmatrix} 0 & 0 \\ \frac{U}{H} & 0 \end{bmatrix} dt = \underline{d\ell(1,0)}$$



THE DISTORTION IS THE COMBINED RESULT OF A ROTATION WITH ANGULAR VELOCITY $-\frac{U}{2H}$ AND A SHEAR STRAIN RATE $\frac{U}{H}$